

GIMPO INTL, Korea, Republic of

WMO: 471100

Lat: 37.558N

Lon: 126.791E

Elev: 58

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.3	11.9	-9.8	3.3	16.2	-4.4	4.3	19.5	19.8	23.8	17.6	26.3	5.4	330	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	91.2	77.4	88.1	75.9	86.1	74.4	80.0	86.0	78.8	84.6	77.5	83.0	8.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.3	83.0	77.1	141.4	81.6	75.5	133.9	80.6	43.7	87.1	42.4	84.1	41.1	83.5	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	15.8	13.9	DB	3.3	94.4	5.0	2.3	-0.3	96.1	-3.3	97.5	-6.1	98.8	-9.8	100.5	(4)
(5)			WB	1.8	81.4	4.8	1.8	-1.6	82.6	-4.4	83.7	-7.1	84.7	-10.6	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	25.8	30.8	41.0	52.9	63.6	72.1	77.7	79.2	70.1	57.6	43.5	30.2		
	DBStd	19.39	7.63	7.19	6.40	6.31	4.57	3.70	3.81	4.44	4.82	6.53	7.86	7.69		
	HDD50	2465	750	537	286	41	0	0	0	0	0	16	221	614		
	HDD65	5339	1215	957	744	365	82	2	0	0	13	238	644	1079		
	CDD50	3863	0	0	8	128	420	662	859	904	602	253	27	0		
	CDD65	1262	0	0	0	2	37	215	394	439	165	10	0	0		
	CDH74	9939	0	0	0	34	317	1337	3142	4063	990	56	0	0		
	CDH80	2965	0	0	0	3	42	312	959	1463	183	3	0	0		
	WSAvg	6.1	5.7	6.1	7.0	6.9	6.6	6.0	6.4	6.1	5.3	5.3	5.8	5.8		
	PrecAvg	56.7	0.7	1.1	1.6	3.6	4.0	5.9	16.8	12.0	6.3	1.9	1.8	0.9		
(15) Precipitation	PrecMax	90.9	2.3	2.5	4.3	6.9	11.1	19.6	41.9	36.9	22.5	8.5	4.2	2.6		
	PrecMin	31.7	0.1	0.0	0.2	1.2	0.5	1.8	6.1	1.6	0.6	0.0	0.3	0.1		
	PrecStd	13.9	0.6	0.8	1.0	1.4	2.2	3.9	8.0	8.7	5.6	1.7	1.0	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	57.0	66.0	78.6	84.3	89.2	93.5	95.2	86.4	79.0	67.9	53.7		
		MCWB	42.7	48.5	51.5	58.1	62.8	68.2	77.8	78.0	72.3	63.9	57.7	46.0		
	2%	DB	44.6	50.2	60.8	73.2	80.5	86.0	89.8	91.8	84.3	75.2	63.0	49.6		
		MCWB	38.0	41.7	48.0	57.4	61.9	68.8	77.2	77.9	70.6	62.3	54.5	43.3		
	5%	DB	41.1	46.5	57.0	69.6	77.3	83.9	87.6	89.7	82.2	71.9	60.6	46.2		
		MCWB	35.3	39.4	46.1	54.2	61.2	68.0	76.4	77.1	69.1	60.4	52.9	41.0		
	10%	DB	37.7	43.0	53.3	65.9	75.0	80.9	85.6	87.6	79.2	69.6	57.2	42.8		
		MCWB	33.0	36.6	43.8	52.8	60.2	67.5	75.1	76.7	67.5	59.0	50.1	38.0		
	0.4%	WB	43.2	50.0	55.0	64.3	68.9	75.6	81.3	82.0	78.5	68.8	60.9	48.2		
		MCDB	47.6	52.7	61.6	70.1	75.5	80.7	86.6	87.9	82.2	73.0	65.0	51.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.1	42.9	50.8	60.0	66.3	73.1	79.7	80.5	75.0	65.5	57.2	44.0		
		MCDB	42.4	48.1	57.1	67.5	73.8	78.7	85.2	86.9	79.8	71.3	61.4	47.8		
	5%	WB	36.3	39.8	47.5	57.1	64.4	71.5	78.4	79.3	72.8	63.3	53.7	41.1		
		MCDB	39.9	45.0	54.0	65.9	72.4	78.1	83.8	85.6	77.6	69.0	59.1	44.9		
	10%	WB	33.6	37.1	44.6	54.7	62.6	70.1	77.1	78.3	70.8	60.9	50.9	38.3		
		MCDB	37.2	41.9	51.1	62.6	70.3	76.6	82.3	84.3	76.2	66.8	55.9	42.3		
	5% DB	MDBR	17.6	18.9	19.6	20.7	19.6	16.8	12.5	13.6	17.3	20.5	18.4	17.0		
		MCDBR	19.3	22.7	24.5	26.1	24.3	21.0	16.1	16.3	18.9	22.0	21.2	20.0		
		MCWBR	14.1	15.1	13.9	12.2	10.3	8.2	6.7	6.2	7.9	11.1	14.1	14.9		
		MCWBR	17.2	19.9	20.8	21.0	18.9	16.0	12.8	13.8	15.1	18.9	18.7	18.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	13.7	14.9	13.8	11.8	10.0	7.7	6.4	6.1	8.4	11.2	14.3	15.1		
(40) Clear-Sky Solar Irradiance	taub		0.437	0.491	0.571	0.618	0.631	0.704	0.672	0.584	0.514	0.506	0.486	0.436		
	taud		1.947	1.853	1.717	1.666	1.690	1.613	1.726	1.931	2.046	1.940	1.931	1.972		
	Ebn at Noon		220	225	220	220	220	204	210	226	234	218	203	210		
	Edh at Noon		44	55	69	77	76	82	73	58	49	50	45	40		
(44) All-Sky Solar Radiation	RadAvg		796	1040	1363	1587	1783	1668	1304	1394	1323	1143	775	694		
	RadStd		49	82	133	108	168	179	200	196	141	79	75	47		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+239	+139
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+90

Nomenclature: See separate page